AP BIOLOGY TOPIC 8.2

Energy Flow Through Ecosystems, Trophic Pyramids & Biogeochemical Cycles

UNIT 8: ECOLOGY (10–15% EXAM WEIGHT)

ENE-1.M | ENE-1.N | ENE-1.O | 10% RULE | NPP = GPP − R

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§1. KEY TERMS & FORMAL DEFINITIONS LEXICON — TOPIC 8.2 (ENERGY FLOW THROUGH ECOSYSTEMS, TROPHIC PYRAMIDS & BIOGEOCHEMICAL CYCLES)

CED Core Vocabulary

Net Energy Gain (Intake > Expenditure) — Excess free energy goes to energy storage, growth, and higher reproductive output .

Net Energy Loss (Expenditure > Intake) — Loss of biomass, halted reproduction, and eventual death.

Asexual vs. Sexual Switching — Organisms like Daphnia (water fleas) and aphids reproduce asexually (fast, low energy cost) when food/conditions are abundant and stable, and switch to sexual...

Autotrophs (Primary Producers) — Form the base of every ecosystem by converting inorganic carbon (CO_2) into organic biomass: Photoautotrophs: Plants, algae, and cyanobacteria capture solar...

Chemoautotrophs — Deep-sea hydrothermal vent prokaryote bacteria oxidize inorganic molecules (H_2S, NH_3, Fe^{2+}) in total darkness without sunlight!

Heterotrophs (Consumers & Decomposers) — Obtain free energy by consuming organic compounds produced by other organisms (herbivores, carnivores, omnivores, scavengers, and decomposers [fungi &...

GPP — Total solar energy captured and converted into chemical energy (glucose) by photosynthesis per unit area per time.

NPP — The remaining energy stored as new plant biomass after producers use R for their own cellular respiration — NPP is the ONLY energy available to primary consumers!

§2. 1. ENDOTHERMS VS. ECTOTHERMS & METABOLIC SCALING

Organismal Bioenergetics

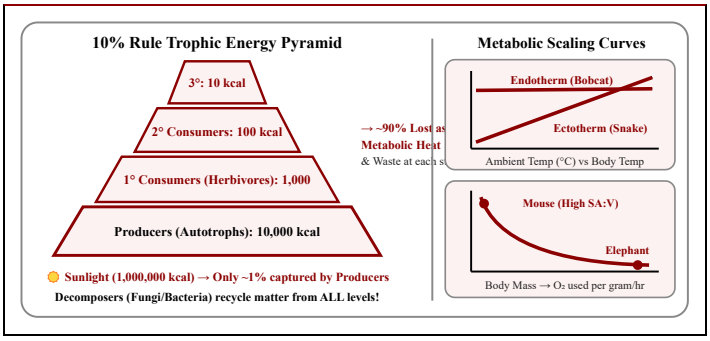
FEATURE	ENDOTHERMS (MAMMALS, BIRDS)	ECTOTHERMS (REPTILES, AMPHIBIANS, FISH, INVERTS)
Heat Source	Generate internal thermal energy via high **cellular respiration / metabolic rate** (plus mitochondrial uncoupling proteins in brown fat).	Rely primarily on **external environmental heat**; regulate body temperature behaviorally (basking vs. shade).
Body Temp & Energy	Maintain **constant internal temperature** across wide ambient temps, but require **much higher food/caloric intake**.	Body temperature & metabolic rate rise/fall with ambient temp; **require far less food energy** (can survive long fasts).

- **Body Size vs. Mass-Specific Metabolic Rate (Surface-Area-to-Volume Ratio):**
  - **Smaller endotherms (e.g., mouse, shrew, hummingbird)** have a **high SA:V ratio**, losing body heat rapidly to the environment → require a **much higher metabolic rate per gram of body tissue** (higher breathing, heart rate, and O_2 consumption/g) than a large elephant!
- **Net Energy Gain vs. Net Energy Loss & Reproductive Strategy:**
  - **Net Energy Gain (Intake > Expenditure):** Excess free energy goes to **energy storage, growth, and higher reproductive output**.
  - **Net Energy Loss (Expenditure > Intake):** Loss of biomass, halted reproduction, and eventual death.
  - **Asexual vs. Sexual Switching:** Organisms like *Daphnia* (water fleas) and aphids reproduce **asexually** (fast, low energy cost) when food/conditions are abundant and stable, and switch to **sexual reproduction** (higher energy cost, generates genetic variation & hardy resting eggs) under environmental stress/crowding!

[COMMON AP EXAM FRQ TRAP] **Ectotherms are NOT "inactive" or "inferior"!** At warm temperatures they are highly active and convert a much higher fraction of ingested food into **new biomass** because they don't burn 80–90% of their calories just to stay warm.

§4. 3. TROPHIC LEVELS & THE 10% RULE OF ECOLOGICAL EFFICIENCY

Energy Flows, Matter Cycles



- **10% Rule of Ecological Efficiency:** On average, only **~10% of the chemical energy** stored in one trophic level is converted into biomass at the next higher trophic level. The remaining **~90% is lost as metabolic heat** (Second Law of Thermodynamics), cellular respiration, movement, and undigested feces.
- **Why Food Chains Are Short (4–5 Trophic Levels Max):** So little energy remains by the 4th/5th level that top-predator populations are small and highly vulnerable to producer decline.

[COMMON AP EXAM FRQ TRAP] **ENERGY FLOWS ONE-WAY; MATTER CYCLES!** Energy enters ecosystems as sunlight and exits as heat (never recycled!). Chemical elements (**C, H, O, N, P**) are conserved and recycled endlessly between biotic and abiotic reservoirs by decomposers.

§5. 4. MAJOR BIOGEOCHEMICAL CYCLES (WATER, C, N, P)

Matter Recycling

CYCLE	MAIN ABIOTIC RESERVOIR	KEY BIOLOGICAL & PHYSICAL PROCESSES
Hydrologic (H_2O)	Oceans (~97%), glaciers, groundwater, atmosphere	Driven by solar energy: **Evaporation, Plant Transpiration, Condensation, Precipitation, Runoff/Percolation**.
Carbon (C)	Atmosphere (**CO_2**), oceans (**HCO_3^-**), fossil fuels, limestone	**Photosynthesis** fixes **CO_2** into organic glucose; **Cellular Respiration, Decomposition, & Fossil Fuel Combustion** return **CO_2** to air.
Nitrogen (N)	Atmosphere (**N_2 gas = 78%** ; triple bond **N≡N** unusable by plants!)	**Prokaryotes required at every step:** (1) **N_2 Fixation** (*Rhizobium* / cyanobacteria: **$N_2 \rightarrow NH_3/NH_4^+$**); (2) **Nitrification** (**$NH_4^+ \rightarrow NO_2^- \rightarrow NO_3^-$**); (3) **Assimilation** by plants; (4) **Ammonification** (decomposers); (5) **Denitrification** (**$NO_3^- \rightarrow N_2$**).
Phosphorus (P)	**Sedimentary Rock & Soil/Water** (*NO atmospheric gas phase!*)	Weathering of rock releases **phosphate (PO_4^{3-})** into soil/water → absorbed by plants (for DNA, RNA, ATP, phospholipids) → consumers → decomposers → sedimentation.

§3. 2. AUTOTROPHS, HETEROTROPHS & PRIMARY PRODUCTIVITY

GPP vs. NPP

HIERARCHICAL CONCEPT & PATHWAY MAP **[CAUSE ➔ MECHANISM ➔ OUTCOME]**

2. Autotrophs, Heterotrophs & Primary Productivity ➔ **Autotrophs (Primary Producers)** ➔ **Heterotrophs (Consumers & Decomposers)** ➔ **GPP** ➔ **NPP**

- **Autotrophs (Primary Producers):** Form the base of every ecosystem by converting inorganic carbon (**CO_2**) into organic biomass:
  - **Photoautotrophs:** Plants, algae, and cyanobacteria capture solar light energy via photosynthesis.
  - **Chemoautotrophs:** Deep-sea hydrothermal vent prokaryote bacteria oxidize inorganic molecules (**H_2S, NH_3, Fe^{2+}**) in total darkness without sunlight!
- **Heterotrophs (Consumers & Decomposers):** Obtain free energy by consuming organic compounds produced by other organisms (*herbivores, carnivores, omnivores, scavengers, and decomposers [fungi & bacteria]*).
- **Gross vs. Net Primary Productivity (GPP vs. NPP):**

NPP = GPP − R (Net Primary Productivity = Gross Primary Productivity − Autotroph Respiration)

- **GPP:** Total solar energy captured and converted into chemical energy (glucose) by photosynthesis per unit area per time.
- **NPP:** The remaining energy stored as **new plant biomass** after producers use **R** for their own cellular respiration — **NPP is the ONLY energy available to primary consumers!**

§6. SCHOLARLY SYNTHESIS: EXPERIMENTAL DESIGN, QUANTITATIVE ANALYSIS, PATHOLOGY & AP FRQ RHETORIC

High-Yield Exam Rigor

MASTER TOPIC SYNTHESIS & CAUSAL ARCHITECTURE — TOPIC 8.2 **[MOLECULAR CAUSE ➔ CELLULAR MECHANISM ➔ PHYSIOLOGICAL / ECOLOGICAL EFFECT ➔ FRQ JUSTIFICATION]**

Topic 8.2: Energy Flow Through Ecosystems, Trophic Pyramids & Biogeochemical Cycles ➔ **Structural / Energetic Determinants** ➔ **Regulated Pathway & Homeostasis** ➔ **Experimental Perturbation (±2 SE_x̄)** ➔ **Phenotypic / Evolutionary Outcome**

SCHOLARLY DIMENSION	COMPREHENSIVE ACADEMIC REFERENCE & MECHANISTIC SPECIFICATION
I. Experimental Design & Required Lab Controls	AP Investigations 10, 11 & 12 (Energy Dynamics, Whole-Plant Transpiration & Pillbug Animal Behavior Choice Chambers): Choice chambers testing pillbug (Armadillidium vulgare) chemotaxis/hydrotaxis/kinesis (count pillbugs on wet vs. dry filter paper every 30 s for 10 min; evaluate using χ^2 against 50:50 null distribution). Potometer/gravimetric transpiration across light, wind, and humidity treatments normalized per cm² leaf surface area. **Null Hypothesis (H_0):** State explicitly that the independent variable treatment has *no statistically significant effect* on the dependent variable; any observed difference is due to random sampling chance alone.
II. Quantitative Formulas & Graph Interpretation	Quantitative Formulas & Ecological Equations: Population Growth Rate: dN/dt = B − D. Exponential Growth: dN/dt = r_max·N (J-curve). Logistic Growth: dN/dt = r_max·N·[(K − N)/K] (S-curve; maximum instantaneous growth rate dN/dt occurs at N = K/2). Simpson's Diversity Index: D_I = 1 − Σ(n / N)² (ranges from 0 to 1; higher value = higher diversity). Net Primary Productivity: NPP = GPP − R. Trophic 10% Rule: ~10% of energy transfers to the next trophic level (~90% lost as metabolic heat/respiration). **Error Bar Rule (±2 SE_x̄):** If the ±2 SE_x̄ error bars of two treatment groups *do not overlap*, there is a statistically significant difference between means ($p < 0.05$; reject H_0).
III. Pathology, Mutation & Disruption Case Study	Ecological Disruption & Trophic Cascade Case Study: Removal of a keystone predator (e.g., Pisaster ochraceus sea star in rocky intertidal zones, or sea otters in kelp forests) triggers a top-down trophic cascade: prey populations (mussels or sea urchins) explode unchecked, competitively excluding other invertebrates or overgrazing kelp producers, collapsing Simpson's Diversity Index. Similarly, agricultural N/P fertilizer runoff causes Cultural Eutrophication: algal blooms block light, aerobic bacteria decompose dead algae and deplete dissolved O_2, creating hypoxic dead zones.
IV. AP FRQ Task Verb Scoring Templates	• **DESCRIBE / IDENTIFY:** State the direct trend, value, or structural component with exact units from the figure. • **EXPLAIN (How/Why):** Connect cause to effect via a 3-link chain: *"[Change in X] causes [molecular/cellular mechanism Y], which results in [observable outcome Z]."* • **PREDICT:** State the specific directional change (increase, decrease, or remain constant) when the pathway is perturbed. • **JUSTIFY:** Pair quantitative data/evidence from the prompt with the underlying biological principle from Topic 8.2 to prove why the claim holds.

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